

# Donghyun Kwon

ASSOCIATE PROFESSOR

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Pusan National University

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## Education

### Seoul National University

PH.D IN DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

- Thesis: Hardware-assisted Isolation for System Framework to Ensure Kernel Integrity
- Advisor: Yunheung Paek
- Distinguished Ph.D. Dissertation Award

Seoul, Korea

Mar 2012 - Feb 2019

### Seoul National University

B.S IN DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

Seoul, Korea

Mar 2008 - Feb 2012

## Professional Experience

### Associate Professor

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING, PUSAN NATIONAL UNIVERSITY

Pusan, South Korea

Mar 2023 - Present

### Assistant Professor

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING, PUSAN NATIONAL UNIVERSITY

Pusan, South Korea

Mar 2020 - Feb 2023

### Senior Researcher

ELECTRONICS AND TELECOMMUNICATIONS RESEARCH INSTITUTE

Daejeon, South Korea

Mar 2019 - Feb 2020

## Publications

\* *corresponding author*

2026

- **Mind the Gap: In-Process Isolation for Safe Rust in WebAssembly**  
Suhyeon Song, Chaewon Shin, and **Donghyun Kwon\***  
European Symposium on Research in Computer Security (ESORICS), 2026
- **XCFI: Comprehensive Control-Flow Integrity for Arm TrustZone-M**  
Yunju Gu, Jaeyeol Park, and **Donghyun Kwon\***  
USENIX Security Symposium, 2026
- **TzPy: Python Runtime for Edge Code Deployment on TrustZone**  
Suhyeon Song, Jaeyeol Park, Chaewon Shin, and **Donghyun Kwon\***  
ACM Transactions on Embedded Computing Systems (TECS), 2026
- **SERA: Secure Micro XRCE-DDS Establishment with Remote Attestation for micro-ROS**  
Jeonghwan Kang, Seungmin Kang, Euijun Kim, Jaeseon Park, Kyoungwan Kim, and **Donghyun Kwon\***  
IEEE Access, 2026
- **WARD: Efficient Memory Protection for WebAssembly on Tiny Embedded Systems**  
Chaewon Shin, Gowon Han, Suhyeon Song, Jinsun Park, Yoon-ho Choi, and **Donghyun Kwon\***  
IEEE Access, 2026
- **Beyond Address Spaces: In-process Memory Isolation for RISC-V**  
Seonghwan Park, Hayoung Kang, and **Donghyun Kwon\***  
Computers & Security, 2026

## 2025

- **SMORE: Practical Redzone-Based Stack Memory Error Detection Mechanism for Embedded Systems**  
Jaeyeol Park, Yunju Gu, and **Donghyun Kwon\***  
Annual Computer Security Applications Conference (ACSAC), 2025
- **ELK: Effective Lock-and-Key Technique for Temporal Memory Safety on Embedded Devices in ARMv8-M**  
Jeonghwan Kang, Kyoungwan Kim, and **Donghyun Kwon\***  
Annual Computer Security Applications Conference (ACSAC), 2025
- **Watch Your Callback: Offline Anomaly Detection using Machine Learning in ROS 2**  
Jeonghwan Kang, Kyoungwan Kim, and **Donghyun Kwon\***  
IEEE Access, 2025
- **WasDom: An Efficient Write Protection for Wasm JITed Code With ARM Domain**  
Suhyeon Song, Chaewon Shin, and **Donghyun Kwon\***  
IEEE Access, 2025
- **ROSec: Intra-Process Isolation for ROS Composition with Memory Protection Keys**  
Jiwon Seo, Kayondo Martin, Jeonghwan Kang, **Donghyun Kwon\***, and Yunheung Paek\*  
IEEE Transactions on Automation Science and Engineering (T-ASE), 2025

## 2024

- **Non-volatile flip-flop with soft error tolerant capability using DICE and C-element**  
Shogo Takahashi, **Donghyun Kwon**, and Kazuteru Namba\*  
IEICE Nonlinear Theory and Its Applications (NOLTA), 2024
- **Look Before You Access: Efficient Heap Memory Safety for Embedded Systems on ARMv8-M**  
Jeonghwan Kang, Jaeyeol Park, Jiwon Seo, and **Donghyun Kwon\***  
Design Automation Conference (DAC), 2024
- **MECAT: Memory-Safe Smart Contracts in ARM TrustZone**  
Seonghwan Park, Hayoung Kang, Sanghun Han, Jonghee M. Youn\*, and **Donghyun Kwon\***  
IEEE Access, 2024

## 2023

- **Enhancing a Lock-and-key Scheme with MTE to Mitigate Use-After-Frees**  
Inyoung Bang, Martin Kayondo, Junseung Yu, **Donghyun Kwon**, Yeongpil Cho\*, and Yunheung Paek\*  
IEEE Access, 2023
- **metaSafer: A Technique to Detect Heap Metadata Corruption in WebAssembly**  
Suhyeon Song, Seonghwan Park, and **Donghyun Kwon\***  
IEEE Access, 2023
- **ZOMETAG: Zone-based Memory Tagging for Fast, Deterministic Detection of Spatial Memory Violations on ARM**  
Jiwon Seo, Junseung You, **Donghyun Kwon**, Yeongpil Cho\*, and Yunheung Paek\*  
IEEE Transactions on Information Forensics and Security (TIFS), 2023
- **SFITAG: Efficient Software Fault Isolation with Memory Tagging for ARM Kernel Extensions**  
Jiwon Seo, Junseung You, Yungi Cho, Yeongpil Cho, **Donghyun Kwon\***, Yunheung Paek\*  
ACM ASIA Conference on Computer and Communications Security (ASIACCS), 2023
- **DEMIX: Domain-Enforced Memory Isolation for Embedded System**  
Haeyoung Kim, Harashta Tatimma Larasati, Jonguk Park, Howon Kim, and **Donghyun Kwon\***  
Sensors, 2023

## 2022

- **Exploring Effective Uses of the Tagged Memory for Reducing Bounds Checking Overheads**  
Jiwon Seo, Inyoung Bang, Yungi Cho, Jangseop Shin, Dongil Hwang, **Donghyun Kwon**, Yeongpil Cho, Yunheung Paek  
The Journal of Supercomputing, 2022
- **Efficient CFI Enforcement for Embedded Systems Using ARM TrustZone-M**  
Gisu Yeo, Yeryeong Kim, Suhyeon Song, and **Donghyun Kwon\***  
IEEE Access, 2022
- **Bratter: An Instruction Set Extension for Forward Control-Flow Integrity in RISC-V**  
Seonghwan Park, Dongwook Kang, Jeonghwan Kang, and **Donghyun Kwon\***

## 2021

- **CoMeT: Configurable Tagged Memory Extension**  
Jinjaee Lee, Derry Pratama, Minjae Kim, Howon Kim, and **Donghyun Kwon\***  
Sensors, 2021
- **A Hardware Platform for Ensuring OS Kernel Integrity on RISC-V**  
**Donghyun Kwon**, Dongil Hwang, and Yunheung Paek  
Electronics, 2021

## 2020

- **RIMI: Instruction-level Memory Isolation for Embedded Systems on RISC-V**  
Haeyoung Kim, Jinjae Lee, Derry Pratama, Asep Muhammad Awaludin, Howon Kim, and **Donghyun Kwon\***  
International Conference on Computer-Aided Design (ICCAD), 2020
- **ProS: Light-weight Privatized Secure OSes in ARM TrustZone**  
**Donghyun Kwon**, Jiwon Seo, Yeongpil Cho, Byoungyoung Lee, and Yunheung Paek  
IEEE Transactions on Mobile Computing (TMC), 2020

## 2019

- **RiskiM: Toward Complete Kernel Protection with Hardware Support**  
Dongil Hwang, Myunghoon Yang, Seongil Jeon, Younghun Lee, **Donghyun Kwon\***, and Yunheung Paek  
Design, Automation and Test in Europe (DATE), 2019
- **Safe and Efficient Implementation of a Security System on ARM using Intra-Level Privilege Separation**  
**Donghyun Kwon**, Hayoon Yi, Yeongpil Cho, and Yunheung Paek  
ACM Transactions on Privacy and Security (TOPS), 2019
- **CRCCount: Pointer Invalidation with Reference Counting to Mitigate Use-After-Free in Legacy C/C++**  
Jangseop Shin, **Donghyun Kwon**, Jiwon Seo, Yeongpil Cho, and Yunheung Paek  
Network and Distributed System Security Symposium (NDSS), 2019
- **uXOM: Efficient eXecute-Only Memory on ARM Cortex-M**  
**Donghyun Kwon**, Jangseop Shin, Giyeol Kim, Byoungyoung Lee, Yeongpil Cho, and Yunheung Paek  
USENIX Security Symposium, 2019

## 2018

- **Hypernel: A Hardware-Assisted Framework for Kernel Protection without Nested Paging**  
**Donghyun Kwon\*\***, Kuenwhhee Oh\*\*, Junmo Park, Seungyong Yang, Yeongpil Cho, Brent Byunghoon Kang, and Yunheung Paek  
Design Automation Conference (DAC), 2018 (\*\* joint first authors)
- **VM-CFI: Control-Flow Integrity for Virtual Machine Kernel Using Intel PT**  
**Donghyun Kwon**, Jiwon Seo, Sehyun Baek, Giyeol Kim, Sunwoo Ahn, and Yunheung Paek  
International Conference on Computational Science and Its Applications (ICCSA), 2018

## 2017

- **Instruction-Level Data Isolation for the Kernel on ARM**  
Yeongpil Cho, **Donghyun Kwon**, and Yunheung Paek  
Design Automation Conference (DAC), 2017
- **Optimization Techniques to Enable Execution Offloading for 3D Video Games**  
**Donghyun Kwon**, Seungjun Yang, Yunheung Paek, and Kwangman Ko  
Multimedia Tools and Applications (MTAP), 2017
- **Dynamic Virtual Address Range Adjustment for Intra-Level Privilege Separation on ARM**  
Yeongpil Cho, **Donghyun Kwon**, Hayoon Yi, and Yunheung Paek  
Network and Distributed System Security Symposium (NDSS), 2017

## 2016

- **Precise Execution Offloading for Applications with Dynamic Behavior in Mobile Cloud Computing**  
Yongin Kwon, Hayoon Yi, **Donghyun Kwon**, Seungjun Yang, Yeongpil Cho, and Yunheung Paek  
Pervasive and Mobile Computing (PMC), 2016

- **Hardware-Assisted On-Demand Hypervisor Activation for Efficient Security Critical Code Execution on Mobile Devices**  
Yeongpil Cho, Jun-Bum Shin, **Donghyun Kwon**, MyungJoo Ham, Yuna Kim, and Yunheung Paek  
USENIX Annual Technical Conference (ATC), 2016

2015

- **Mantis: Efficient Predictions of Execution Time, Energy Usage, Memory Usage and Network Usage on Smart Mobile Devices**  
Yongin Kwon, Sangmin Lee, Hayoon Yi, **Donghyun Kwon**, Seungjun Yang, Byung-gon Chun, Ling Huang, Petros Maniatis, Mayur Naik, and Yunheung Paek  
IEEE Transactions on Mobile Computing (TMC), 2015

2014

- **Techniques to Minimize State Transfer Costs for Dynamic Execution Offloading in Mobile Cloud Computing**  
Seungjun Yang, **Donghyun Kwon**, Hayoon Yi, Yeongpil Cho, Yongin Kwon, and Yunheung Paek  
IEEE Transactions on Mobile Computing (TMC), 2014

2013

- **Fast Dynamic Execution Offloading for Efficient Mobile Cloud Computing**  
Seungjun Yang, Yongin Kwon, Yeongpil Cho, Hayoon Yi, **Donghyun Kwon**, Jonghee Youn, and Yunheung Paek  
IEEE International Conference on Pervasive Computing and Communications (PerCom), 2013
- **Mantis: Automatic Performance Prediction for Smartphone Applications**  
Yongin Kwon, Sangmin Lee, Hayoon Yi, **Donghyun Kwon**, Seungjun Yang, Byung-Gon Chun, Ling Huang, Petros Maniatis, Mayur Naik, and Yunheung Paek  
USENIX Annual Technical Conference (ATC), 2013

## Academic Service

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- (Reviewer) ACM Transactions on Privacy and Security (TOPS)
- (Reviewer) ACM Transactions on Architecture and Code Optimization (TACO)
- (Reviewer) IEEE Transactions on Information Forensics & Security (TIFS)
- (Reviewer) IEEE Transactions on Computers (TC)
- (Reviewer) IEEE Access
- (Reviewer) Journal of Supercomputing
- (Associate Editor) Journal of Information Processing Systems (JIPS)

## Teaching

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- Grad. **System Vulnerabilities and Countermeasures**, Fall 2021 - Fall 2025
- Grad. **Introduction to System and Software Security**, Spring 2021 - Spring 2026
- Under. **Web Application Programming**, Fall 2021 - Fall 2023
- Under. **Logic Circuit Design & LAB.**, Fall 2023 - Fall 2024
- Under. **Compiler**, Fall 2022
- Under. **Computer Architecture**, Fall 2024 - Fall 2025
- Under. **Blockchain**, Spring 2024 - Spring 2026
- Under. **Programming Principles and Practice**, Fall 2020 - Fall 2021
- Under. **Adventure Design**, Fall 2020
- Under. **Introduction to Internet and Web**, Spring 2020 - Spring 2023
- Under. **System Software**, Spring 2020 - Spring 2026
- Under. **Discrete Mathematics 1**, Spring 2020